

MetalSynth-Pelvis: Conditioned synthesis of osteosynthesis implants and local metal artifacts in pelvic CT scan volumes using Multi-Window Latent Diffusion

Kiara Alexandra Balcázar Santa Cruz — UTEC · Advisor: Víctor Flores Benites

1. Problem Statement

Automatic pelvic computed tomography (CT) scan analysis and volumetric segmentation relies heavily on the availability of sufficiently annotated volumetric data of sufficient size. However, few real-world examples of annotated CTs with embedded metallic osteosynthesis materials (i.e. screws, plates, etc.) are found in any annotated corpus, giving rise to a severe domain shift issue for medical volumetric analysis. For instance, the largest published pelvic bone volumetric annotation set CTPelvic1K (Liu et al., 2021) offers a total of 1,184 volumes, of which, however, 14/1,184 exhibit a supervised annotation in the bone (only), while only 75/1,184 contain metal at all. As a result, existing state-of-the-art deep learning segmentation models are trained exclusively on "metal-free" anatomy, and present a dramatic drop in performance in the so-called peri-implant zone, a local region where local metal artifacts (beam hardening, streaking, photon starvation) hinder the clarity of anatomical boundaries. Ironically, it is the peri-implant region where actual clinical conclusions are crucial, for example the detection of periprosthetic fractures or loosening of the implant.

While generative models based on diffusion processes excel at synthesizing visually plausible medical anomalies, they structurally fail when applied to metallic implants. State-of-the-art lesion synthesis models, such as DiffTumor (Chen et al., 2024), DiffBoost (Zhang et al., 2025), CLAIM (Ramzan et al., 2026), and LGESynthNet (Jacob et al., 2026), are designed around bounded inpainting for deformable biological tissues. They implicitly assume non-rigidity and strictly restrict intensity alterations to the inside of the object's mask. This is physically incompatible with metal implants, which are rigidly geometric and produce severe attenuation profiles (beam hardening, photon starvation, and high-amplitude streaking) that propagate globally far beyond the implant's boundaries, destroying the signal of the surrounding soft tissue.

Conversely, physics-based simulations for Metal Artifact Reduction (MAR) achieve perfect physical coherence but completely lack scalable generative placement. The gold-standard physics simulator, XCIST/CatSim (De Man et al., 2007; Wu et al., 2022), parameterized by the AAPM CT-MAR challenge framework (Peters et al., 2025), accurately propagates polychromatic beam scatter through the volume. However, its use as a mass data-augmentation tool is bottlenecked by its reliance on the manual, expert-driven insertion of virtual implants into the anatomy. Other projection-domain insertion methods (Ren et al., 2022; Wang et al., 2019; Singhrao et al., 2024) similarly lack an automated, anatomy-aware placement engine, making them unscalable for training deep learning models.

Even if automated placement were achieved, mere anatomical containment is insufficient; a valid simulation must reflect true surgical admissibility. Traditional deterministic surgical planning pipelines (Liu et al., 2025) attempt to optimize a single "perfect" trajectory. Yet, real-world clinical data reveals a different reality: Zwiggmann et al. (2009) established that 31–60% of real-world iliosacral screw placements result in some degree of malposition or cortical breach, based on the clinical scale of Smith et al. (2006). Training a diagnostic network solely on "perfect" deterministic trajectories will inevitably cause it to fail when encountering the true variance of clinical errors. Therefore, a generative sampler must not optimize for perfection, but instead sample from the true statistical distribution of clinical malpositions, bounded by operative safe zones (Ramadanov and Zabler, 2025) and pelvic density representations (Arand et al., 2019).

Recognized Gap. Current attempts to bridge the metal-artifact domain shift are bottlenecked by three mutually exclusive paradigms: physical simulators lack scalable automated placement, biological diffusion models cannot generate rigid global streaks, and deterministic surgical planners ignore the statistical reality of clinical malposition. To the best of the author's knowledge, there is no diffusion-based method that overcomes these barriers by combining: **(C1)** Non-biological rigid implant geometries (compensating for thresholding over-coverage bias, Xie et al., 2024); **(C2)** A surgically constrained placement sampler that accurately emulates the population distribution of clinical malpositions; and **(C3)** A multi-window latent diffusion synthesis module (Wang et al., 2025) that extends the generation band (B_δ) beyond the implant mask to explicitly model global streaking and beam hardening. This is the gap this work aims to fill.

2. Research Question

What improvement does a surgical admissibility-constrained (bone density and cortical containment) 3D placement sampler, a multi-window Latent Diffusion Model (LDM) conditioned on the rigid implant geometries and an extended generation band (B_δ), have in terms of distributional physical-coherence of the synthesized pelvic osteosynthesis implants and their local artifacts and the corresponding downstream impact on bone volumetric segmentation networks on metal artifact-affected CTs, when compared with both naive copy-paste insertion and the explicit physics-based XCIST/CatSim simulator.

3. Hypothesis

The synthetic generation of medical anomalies and metallic implants using a rigid geometry and extended spatial boundaries, HU-conditioned diffusion model and whose placement is surgically enforced via a density-based placement sampler is more physically-coherent than unconstrained generative baseline methods and on the same footing as an explicit physical simulation (XCIST/CatSim). This physical-coherence, quantified by novel Surgical Admissibility of Placement (SAP) and Boundary Feature Coherence (BFC) metrics, emulates the true clinical distribution of real-world implant placement. Subsequently, the downstream peri-implant bone volumetric segmentation (Dice, HD95) using this type of synthetic data as a training augmentation improves downstream on CTPelvic1K's test set, with full ablations revealing the non-redundant role of the multi-window encoding and extended boundary generation.

4. Objectives

General Objective. Design, implement and evaluate a physics-and-anatomy-aware latent diffusion pipeline with rigid surgical constraints and extended HU-based conditioning, capable of generating physically coherent metallic implants for the improved robustness of downstream segmentation networks.

1. **Validation of a multi-window representation (Compulsory Go/No-Go):** Ensure HU reconstruction error (HU to multi-window to VAE to HU again) within the bone tissue below a 25 HU threshold (Mean Absolute Error (MAE)). This will serve as a guarantee that the intensity gradients that drive downstream applications are preserved.

2. **Design of surgically constrained placement sampler:** propose position-orientation 3D pose constrained by bone density maps (Arand et al., 2019), cortical boundaries, and corridor continuity that results in a distribution of placements that emulates the true clinical 31–60% real-world imperfection rate, rather than a single artificial "optimized" trajectory.
3. **Design of a conditioned synthesizer:** 2.5D ControlNet-guided Latent Diffusion Model (Stable Diffusion 1.5 backbone). A crucial detail in the formulation is that the synthesis mask must be extended with a spatial band B_δ ($\sim 12\text{mm}$) over the implant boundary, to allow a more explicit manifestation of the streaking artifacts and beam hardening that occur outside of the implant mask in the generated image.
4. **Formalization of physical-coherence metrics:** Formulation of SAP and BFC and Inter-slice Consistency (ISC) to decouple the evaluation of the placement sampler from the appearance synthesizer, using Wasserstein-1 distances with the real-world clinical distributions.
5. **Evaluation of the downstream impact:** Segmentation improvement, using established metric (Dice, HD95), on real metal-impaired cases against comprehensive baselines, including the physics-based XCIST/CatSim, which is verified with full per-constraint ablations.

5. Expected Results

Goal	Metric(s)	Expected Outcome
Surgical Admissibility (Novel Sampler Eval)	SAP: Cortical Breach Grade (Smith et al., 2006); Density Zone fraction.	Distribution matching: Wasserstein-1 distance reveals that generated implant placements are close to the 31-60% clinical real-world malposition rate, rather than unrealistic "perfect" optimization.
Appearance Coherence (Synthesizer Eval)	BFC: Boundary Feature Coherence. ISC: Inter-slice Consistency. Streak Amplitude: vs real.	Lower profile discrepancy than naive copy-paste insertion, with streak amplitudes showing a statistically comparable, or better, result than the XCIST/CatSim explicit physics baseline.
Representation Viability (Standard Encoding Eval)	MAE in HU (Round-trip)	Error in the bone tissues strictly below 25 HU threshold. Log-compressed metal ranges preserve the structural topology.
Downstream Detection (Clinical Utility)	Dice, HD95 (Peri-implant restricted)	Outperforms conventional unconstrained diffusion augmentation on CTPelvic1K clinical evaluation set.
Ablation validation (Constraint Verification)	All metrics above	Incremental ordering shows their non-redundant contribution. Degradation when B_δ band and multi-window HU encoding are removed.

6. Datasets

Primary Dataset. CTPelvic1K (Liu et al., 2021) will be used as the only source of patient anatomy and true artifact distributions. It consists of a total of volumes, while the **CLINIC-metal** subset regroups the studies with metal at all. *Strict Isolation Rule.* No case can act at the same time as training (geometry bank/appearance prior) and as a evaluation function. The geometry of test implants will never reach the bank consumed by the proposed sampler or the physical simulation arm. Thus, total prevention of data leakage.

Secondary/validation dataset. AAPM CT-MAR Challenge Data (Peters et al., 2025; Haneda et al., 2025). The 14,000 public challenge cases will never be used for the training of any models. They will be used solely to check the fidelity of the baseline reimplementaion (XCIST/CatSim). The challenge simulation must hit $\text{RMSE} \leq 30 \text{ HU}$ and $\text{SSIM} \geq 0.95$ on published challenge data for it to be approved for application on CTPelvic1K as a comparison baseline.

7. References

- Arand, C., et al. (2019). 3D Statistical Model of the Pelvic Ring — A CT-Based Statistical Evaluation of Anatomical Variation. *Journal of Anatomy*, 234(3):376–383.
- Chen, Q., et al. (2024). Towards Generalizable Tumor Synthesis. *CVPR 2024*, pp. 11147–11158.
- De Man, B., et al. (2007). CatSim: A New Computer Assisted Tomography Simulation Environment. *SPIE Medical Imaging*, 65102G.
- Haneda, E., et al. (2025). AAPM CT Metal Artifact Reduction Grand Challenge. *Medical Physics*, 52(10):e70050.
- Jacob, A. J., et al. (2026). LGEsSynthNet: Controlled Scar Synthesis for Improved Scar Segmentation in Cardiac LGE-MRI Imaging. *STACOM 2025*, LNCS vol. 16459.
- Karageorgos, G. M., et al. (2024). A Denoising Diffusion Probabilistic Model for Metal Artifact Reduction in CT. *IEEE TMI*, 43(10):3521–3532.
- Kazerouni, A., et al. (2023). Diffusion Models in Medical Imaging: A Comprehensive Survey. *Medical Image Analysis*, 88:102846.
- Liu, J., et al. (2025). An End-to-End Geometry-Based Pipeline for Automatic Preoperative Surgical Planning of Pelvic Fracture Reduction and Fixation. *IEEE TMI*, 44(1):79–91.
- Liu, P., et al. (2021). Deep Learning to Segment Pelvic Bones: Large-Scale CT Datasets and Baseline Models. *Int J CARS*, 16:749–756.
- Peters, N., et al. (2025). A Hybrid Training Database and Evaluation Benchmark for Assessing Metal Artifact Reduction Methods for X-ray CT Imaging. *Medical Physics*, 52(10):e70020.
- Ramadanov, N., Zabler, S. (2025). Ramadanov–Zabler Safe Zone for Sacroiliac Screw Placement: A CT-Based Computational Pilot Study. *Journal of Clinical Medicine*, 14(10):3567.
- Ramzan, F., et al. (2026). CLAIM: Clinically-Guided LGE Augmentation for Realistic and Diverse Myocardial Scar Synthesis and Segmentation. *AliH 2025*, LNCS vol. 16038.
- Ren, L., et al. (2022). Framework of Metal Insertion in the Projection Domain for Image Quality Optimization in Interventional Computed Tomography. *Journal of Medical Imaging*, 9(3):035001.
- Rombach, R., et al. (2022). High-Resolution Image Synthesis with Latent Diffusion Models. *CVPR 2022*, pp. 10674–10685.
- Selles, M., et al. (2024). Advances in Metal Artifact Reduction in CT Images: A Review of Traditional and Novel Metal Artifact Reduction Techniques. *European Journal of Radiology*, 170:111276.
- Singhrao, K., et al. (2024). End-to-End Validation of Fiducial Tracking Accuracy in Robotic Radiosurgery Using MRI-Only Simulation Imaging. *Medical Physics*, 51:31–41.
- Smith, H. E., et al. (2006). An Evaluation of Image-Guided Technologies in the Placement of Percutaneous Iliosacral Screws. *Spine*, 31:234–238.
- van Bosse, H. J. P., et al. (2011). Pelvic Positioning Creates Error in CT Acetabular Measurements. *CORR*, 469:1683–1691.
- Wang, H., et al. (2025). Adaptive Weighting Based Metal Artifact Reduction in CT Images. *IEEE TMI*, 44(6):2408–2423.
- Wang, Z., et al. (2019). Deep Learning Based Metal Artifacts Reduction in Post-operative Cochlear Implant CT Imaging. *MICCAI 2019*, LNCS vol. 11769.
- Wu, M., et al. (2022). XCIST — An Open Access X-ray/CT Simulation Toolkit. *Physics in Medicine & Biology*, 67(19):194002.
- Xie, K., et al. (2024). Metal Implant Segmentation in CT Images Based on Diffusion Model. *BMC Medical Imaging*, 24:204.
- Yun, S., et al. (2026). A Strategy for Simulation-Driven CT Metal Artifact Reduction Toward Improving Network Generalizability. *Medical Physics*, 53:e70336.
- Zhang, L., et al. (2023). Adding Conditional Control to Text-to-Image Diffusion Models. *ICCV 2023*, pp. 3813–3824.
- Zhang, Y., et al. (2026). Enhancing Intraoperative Pedicle Screw Planning Accuracy via Diffusion-Based Synthetic CT. *Journal of X-Ray Science and Technology*, 34(4):525–540.
- Zhang, Z., et al. (2025). DiffBoost: Enhancing Medical Image Segmentation via Text-Guided Diffusion Model. *IEEE TMI*, 44(9):3670–3682.
- Zwingmann, J., et al. (2009). Computer-Navigated Iliosacral Screw Insertion Reduces Malposition Rate and Radiation Exposure. *CORR*, 467(7):1833–1838.